Marginal-Price Support for Electric-Bus Fleets: An Exact Assignment-Dependent Projection and Its Price Dynamics

Nathan Cho

Draft, 30 October 2026

Abstract

An electric-bus operator large enough to move the electricity price faces a feedback loop: prices shape the schedule, the schedule shapes the load, and the load shapes the price. Because a bus cannot cover a fraction of a service, the operator's set of attainable loads and costs is nonconvex, so posted marginal prices may fail to support any physically executable schedule. We characterize this failure exactly for a stylized but operationally recognizable class: morning and afternoon service batches, a shared midday charging window, overnight replenishment, and any allowed set of morning–afternoon pairings. Our main theorem shows that, for every number of paired services, a single minimum-requirement matching determines the complete joint set of grid loads and vehicle costs, a polygon with at most five vertices, including charging losses and vehicle-to-grid exports. On this exact description we prove three results about price formation. A positive support gap requires the marginal-cost spread between charging windows to reach the daily vehicle cost divided by the largest forced-charging increment, and it is at most one eighth of the spread's slope times that increment squared. With a small constant step, price iteration converges when the hull optimum is a vertex of the cost frontier and never converges when it lies inside a face. A face optimum occurs with or without a support gap, so a price cycle is not evidence of nonconvexity. Diminishing steps converge to the convex-hull price, and with quadratic supply costs their time-averaged schedules recover the convex-hull plan. Edge-dependent transition energy and a third service batch mark where the reduction ends. A study of 18 German bus operators, using public timetable data, shows that with 400 kWh batteries, for 15 operators every morning–afternoon pairing needs no midday charging, so support cannot fail at any price. Wherever failure is possible, it needs a spread of at least 0.5 EUR/kWh, and more than 1 EUR/kWh for the median operator. Undamped price iteration nonetheless cycles in 15% of the scenarios, almost all of them without a support gap.

1 Introduction

Electrified bus depots are becoming some of the largest flexible loads on urban distribution networks. A depot's charging schedule is no longer a price-taker's afterthought: its load can move a local marginal cost, a congestion charge or a negotiated tariff, and that price in turn changes the schedule. Classical electric-vehicle scheduling treats electricity prices as exogenous [6, 7, 19, 24]. Once prices respond to the fleet, two questions arise. Is there a posted price at which the operator's own cost-minimizing schedule is the system-optimal one? And if prices are adjusted iteratively in response to the fleet's load, do they settle?

The obstacle is indivisibility. A bus serves a whole service or none of it, so the set of loads and costs the operator can attain is a union of pieces. When the system-optimal load lies between

pieces, no posted marginal price can induce it: the operator either over- or under-reacts. This is the familiar nonconvex-pricing problem of electricity markets [9, 11, 18, 20], but here the nonconvexity is created by *service assignment*. Pairing two services on one bus saves a vehicle but may force that bus to recharge between them. A fixed-population charging aggregate, however fine-grained [14–16], does not retain that trade-off because the population of vehicle duties is itself a decision.

This paper gives an exact account of that trade-off for a class that is simple enough to solve and recognizable enough to matter: buses serve a morning peak batch and an afternoon peak batch, may charge at the depot in a common midday window, and must be full again after an overnight window. Any bus may serve at most one morning and one afternoon run, and a declared compatibility graph lists the allowed pairings.

Contributions.

1. **Exact projection (Theorem 3).** For each matching cardinality m, all assignments with m pairs share every aggregate quantity except one: the total early energy their paired buses must receive. A matching minimizing that requirement contains the *entire* physical load–cost polygon of every other matching of the same size, at identical loads and fees. The complete projection is therefore a union of at most $\nu+1$ polygons with at most five vertices each, with common charging and discharging losses, a shared bidirectional connector and full overnight replenishment. Each point has a finite, serial, state-of-charge-feasible realization.

2. **Anatomy of the support gap (Proposition 6).** In the lossless case the gap $\Delta = D - \mathrm{CH}$ between physical and convexified planning values is positive only if the marginal-cost spread between the two charging windows reaches f/η, where f is the daily vehicle cost and η an increment between consecutive forced-charging thresholds. The gap is at most $\kappa\eta^2/8$, where κ bounds the slope of that spread.

3. **Price dynamics (Theorem 8).** For the damped price-adjustment loop that posts the marginal cost of the last response, undamped prices are eventually periodic with period at most two and can cycle even when a supporting price exists. With a small constant step, prices converge if the hull optimum is a vertex of the cost frontier, and never converge if it lies in the relative interior of a face. In the latter case the loads alternate between the face's two endpoints and the prices stay within an $O(a)$ band around the convex-hull price. Diminishing steps converge to that price, and with quadratic supply costs the step-weighted time averages of the loads recover the hull optimum. A face optimum arises both with and without a support gap. Persistent cycling therefore diagnoses a degenerate tangency, not nonconvexity.

4. **Where the reduction stops (Section 6).** A common transition energy per pair preserves the reduction. Edge-dependent transition energy breaks it: on a complete assignment graph where every perfect matching is feasible, a strongly convex supply cost still leaves a positive gap. With a third service batch, fleet feasibility alone is strongly NP-complete. On one public 37-service timetable outside the two-batch class, rational certificates establish a strictly positive gap.

5. **Calibrated evidence (Section 7).** Using the public timetables of 18 German operators [21, 22], we build morning and afternoon peak runs and compute when the obstruction can arise. With 400 kWh batteries, for 15 of 18 operators every morning–afternoon pairing needs no midday charging, and support failure is impossible at any price. Across battery sizes, failure needs a marginal-cost spread of at least 0.5 EUR/kWh for a 150 EUR/day vehicle cost, and more than 1 EUR/kWh for the median operator. When it occurs, the gap is below 0.06% of total daily cost. Undamped price iteration nevertheless cycles in 15% of scenarios, nearly all of which have no gap.

The support criterion itself, matching integrality and convex-hull pricing are classical tools, and we use them as such. The new content is the exact physical description of the assignment-dependent load–cost set, together with the consequences it makes explicit for support and price iteration. The model is deliberately stylized: it abstracts from travel between services, staggered availability, battery degradation and switching time. Section 6 shows which of these abstractions are essential.

Organization. Section 2 positions the work. Section 3 defines complete plans and support. Section 4 proves the projection theorem, Section 5 the price-formation results, and Section 6 the limits of the reduction. Section 7 reports the calibrated study, and Section 8 discusses implications. Proofs are in the appendices; code and data reproducing every number and figure are in the accompanying repository (Appendix F).

2 Related work

Electric-bus scheduling with charging. Exact and heuristic methods for the electric vehicle scheduling problem combine service covering with charging decisions under station capacity [7], scale through column generation [19], or integrate flow and set-partitioning models [6]. Wu et al. [24] add power-grid characteristics and endogenous vehicle-to-trip assignment with overnight charging. These works optimize schedules at given prices or grid limits; they do not characterize the set of loads and vehicle costs attainable across assignments, which is what price support requires.

Buses, tariffs and markets. Wu et al. [25] embed bus operations in a bi-level model with distribution locational marginal prices. Their upper level is a tactical frequency-setting problem with terminal opportunity charging; the lower level clears a DC power-flow market, and the bi-level problem is solved through its complementarity reformulation. Vehicle-to-service assignment is not modeled, so the assignment nonconvexity studied here does not arise. Zoltowska and Lin [27] aggregate bus charging into day-ahead auction bids; Blatiak et al. [5] optimize commercial-fleet trip timing and charging against energy and balancing prices; and Zhou et al. [26] formulate a multi-depot mixed-integer charging and discharging schedule for electric buses under vehicle-to-grid tariffs and solve it with an adaptive large-neighbourhood search (our comparison with the last is limited to its published abstract). All of these optimize a richer operational model. None gives an exact description of the assignment-dependent load–cost set or analyses support and price iteration.

Aggregate flexibility. Exact characterizations of the flexibility of a given EV population use permutahedra and generalized polymatroids [14, 15]; Müller et al. [16] aggregate and disaggregate costs for zonotopic resource sets; Al Taha et al. [2] and Al Taha and Bitar [1] treat lossless and lossy storage aggregation; and Arvis et al. [4] integrate aggregated EV flexibility into unit commitment. For one fixed matching, our early-energy interval is an elementary specialization of such aggregation. The distinct step is the *cross-assignment* identity: across all matchings of one cardinality, the union of their exact polygons equals the polygon of one minimum-requirement matching. Generic disjunctive formulations can encode that union, but they do not establish this replacement, and a net-interval aggregate alone loses the mandatory-purchase statistic that drives it (Remark 4).

Nonconvex pricing and price adjustment. Uplift, convex-hull pricing and their computation are established [3, 9, 11, 12, 18, 20]; the Shapley–Folkman lemma explains why aggregate nonconvexity is small in large markets [10, 23]. Our support criterion (Proposition 2) is the standard

one specialized to a complete-fleet participant. Iterative price adjustment is a Lagrangian-dual method; for convex programs, ergodic averages of subproblem solutions recover primal optima [13, 17]. Theorem 8 is a one-dimensional, nonconvex counterpart. The response is set-valued and piecewise constant, the update uses the supply's marginal cost rather than a dual gradient, and the conclusions are finite-time statements about which physical schedules the iteration visits.

3 Complete fleets and the meaning of support

Fix mandatory services, movement possibilities, a charging horizon divided into market intervals $t = 1, \dots, T$, and physical equipment. A *complete plan* $x \in \mathcal{X}$ covers every mandatory service exactly once, assigns services to used buses, routes them, and schedules charging subject to battery and shared-resource limits. Let $e(x) \in \mathbb{R}^T$ be the fleet's net grid energy by interval (negative for exports) and $c(x)$ its intrinsic operating cost, here the vehicle cost. The projected image $S = \{(e(x), c(x)) : x \in \mathcal{X}\}$ is assumed nonempty and compact.

A supplier serves the fleet's incremental demand at a proper closed convex cost $F : \mathbb{R}^T \to \mathbb{R} \cup \{+\infty\}$, finite on all attainable loads. The *physical* and *convexified* planning values and the *gap* are

$$D = \min_{x \in \mathcal{X}} \{c(x) + F(e(x))\}, \qquad \mathrm{CH} = \min_{(e,c) \in \operatorname{conv} S} \{c + F(e)\}, \qquad \Delta = D - \mathrm{CH} \geq 0. \tag{1}$$

The hull mixes complete plans. Its optimum is a schedule mixture, not one executable day. At a posted price vector $p \in \mathbb{R}^T$ the fleet's response value and the regret of plan x are

$$V(p) = \min_{y \in \mathcal{X}} \{c(y) + p^\top e(y)\}, \qquad r(x; p) = c(x) + p^\top e(x) - V(p). \tag{2}$$

Definition 1 (Own-load support)**.** A physical plan x is *supported* if some marginal price $p \in \partial F(e(x))$ gives $r(x; p) = 0$.

The operator in (2) controls the complete fleet and takes the posted price as fixed. This is the decentralization test behind marginal-cost tariffs: a price computed from the operator's own load must not give it a reason to deviate. A strategic operator anticipating the price change would solve a different problem, which we do not study.

Proposition 2 (Classical support criterion)**.** *For every physical x and $p \in \partial F(e(x))$, $r(x; p) \geq c(x) + F(e(x)) - \mathrm{CH} \geq \Delta$. If $\operatorname{relint} \operatorname{conv} e(\mathcal{X}) \cap \operatorname{relint} \operatorname{dom} F \neq \emptyset$, then some physical plan is supported if and only if $\Delta = 0$, and in that case every physical optimum is supported by every maximizer of $V(p) - F^*(p)$.*

This specializes standard arguments [9, 11, 18]; the proof is in Appendix B. Two consequences are used below. A positive gap gives every physical plan a regret of at least Δ at its own marginal prices. Conversely, at a maximizer p^* of $V(p) - F^*(p)$ (a convex-hull price), the identity in the proof shows that a physical optimum's regret is at most Δ, so an uplift payment of at most Δ supports it.

4 An exact assignment-dependent projection

The two-batch class. There are $n_A \geq 1$ morning (A) services during $[0, 1]$ and $n_B \geq 1$ afternoon (B) services during $[2, 3]$, with battery draws $0 < a_i, b_j \leq B$ for a common usable capacity $B > 0$. Same-batch services overlap, so each bus serves at most one A and one B service. A bipartite

graph Γ lists the allowed A–B pairings; every service may also be served by its own bus. A partial matching $M \subseteq \Gamma$ with $|M| = m$ uses $N_m = n_A + n_B - m$ buses, each at cost $f > 0$; let ν be the maximum matching size. Transitions take no time or energy (Section 6 relaxes this). All buses start full, are at the depot throughout the *early* window $[1, 2]$, and must be full again after the *terminal* window $[3, 4]$. One shared early connector moves at most $K \geq 0$ grid-side energy in either direction; one terminal connector delivers at most $H \geq 0$, charging only. Charging and discharging efficiencies are common, $\alpha, \delta \in (0, 1]$. Early actions may change direction finitely often, are sequential on the connector, and incur no switching delay or cycling cost.

Let C and X be the integrated early purchases and exports. The connector's gross use is $C + X \leq K$ and the net early load is $E = C - X$. Write

$$S_0 = \sum_i a_i + \sum_j b_j, \qquad P = \sum_i a_i, \qquad Q = \alpha C - X/\delta.$$

Q is the fleet's net battery gain in the early window. Full replenishment then requires the terminal purchase $L = (S_0 - Q)/\alpha$. The cost F is a function of the two window loads (E, L).

Theorem 3 (Complete common-window projection)**.** *For $m = 0, \ldots, \nu$ let*

$$r_m = \min_{M \subseteq \Gamma,\, |M| = m} \sum_{(i,j) \in M} (a_i + b_j - B)_+,$$

$$W_m = \Big\{(C, X) :\ C \geq r_m/\alpha,\ X \geq 0,\ C + X \leq K,\ \max\{S_0 - N_m B,\ S_0 - \alpha H\} \leq \alpha C - X/\delta \leq P\Big\}.$$

The set of attainable (early load, terminal load, vehicle cost) triples is exactly

$$\mathcal{P} = \bigcup_{m=0}^{\nu} \Big\{(C - X,\ (S_0 - \alpha C + X/\delta)/\alpha,\ f N_m) : (C, X) \in W_m\Big\}.$$

Every point has a realization with finitely many serial actions. Each nonempty W_m is a polygon with at most five vertices, so conv $\mathcal{P}$ *is generated by at most $5(\nu+1)$ points. If exports are prohibited, the same holds with $X = 0$.*

Why one matching statistic suffices. Fix a matching. A bus's net early battery gain x_v lies in an interval: $[a_i + b_j - B,\ a_i]$ for a paired bus, $[a_i - B,\ a_i]$ for an A-only bus and $[b_j - B,\ 0]$ for a B-only bus. The lower endpoints sum to $S_0 - N_m B$ and the upper endpoints to P, whatever the matching. Only the sum R of the *positive* lower endpoints, the energy that paired buses must receive before their afternoon service, depends on which services are paired. At any admissible total Q, the intervals can be filled so that positive gains total $\max\{R, Q\}$, giving one-direction purchases $\max\{R, Q\}/\alpha$ and exports $\delta(\max\{R, Q\} - Q)$. Any additional purchase $z \geq 0$ with matching exports $\alpha\delta z$ is realized by battery-neutral charge–discharge cycles on one bus, in chunks of depth at most $B/2$. Serializing all actions uses exactly $C + X$ of connector capacity. At fixed m, lowering R only relaxes the constraint $C \geq R/\alpha$, so a minimum-requirement matching contains the polygon of every other matching of that size. This is set equality, not just economic dominance; the complete proof is in Appendix A.

Remark 4 (What a net-interval aggregate loses). Two fleets whose battery-gain intervals sum to the same net interval can have different mandatory purchases: $[1, 2]$ and $[-1, 0]$ sum to $[0, 2]$, as do $[0, 1]$ and $[0, 1]$, but the first pair forces at least one unit of purchase. With losses and a gross connector budget this difference changes the attainable (E, L) set. The statistic r_m is exactly the information about the assignment that the aggregate must keep.

The lossless charge-only case. With $\alpha = \delta = 1$ and no exports, each layer reduces to an interval of early loads. Let $\mathrm{Lo} = \max\{0, S_0 - H\}$, $U = \min\{P, K\}$ and $\ell_m = \max\{\mathrm{Lo}, r_m\}$. Then

$$\mathcal{P} = \bigcup_{m:\ \ell_m \le U} \{(E,\ S_0 - E,\ fN_m) :\ \ell_m \le E \le U\}, \tag{3}$$

the model is feasible exactly when $\mathrm{Lo} \le U$, and r_m is nondecreasing and discretely convex in m (Appendix A). Let m_0 be the largest m with $r_m \le \mathrm{Lo}$ and $m_{\max}$ the largest m with $\ell_m \le U$. The lower boundary $\bar{c}(E)$ of the hull's operating cost is piecewise linear. Call Lo and r_m for $m_0 < m \le m_{\max}$ the *breakpoints*. The boundary starts at (Lo, fN_{m_0}), interpolates linearly between consecutive breakpoints (ℓ_{m-1}, fN_{m-1}) and (ℓ_m, fN_m), and is flat at $fN_{m_{\max}}$ from the last breakpoint to U (the *flat face*). Consecutive breakpoints are one cardinality apart, so each sloping segment has slope $-f/\eta_m$ with increment $\eta_m = \ell_m - \ell_{m-1} > 0$.

Corollary 5 (Support classification)**.** *Let $G(E) = F(E, S_0 - E)$ be continuous and strictly convex on $[\mathrm{Lo}, U]$ and let E^* minimize $\bar{c} + G$. Then $\Delta = 0$ if and only if E^* is physical: E^* is a breakpoint or lies in the flat face $[\ell_{m_{\max}}, U]$. The gap is positive exactly when E^* lies strictly between two consecutive breakpoints.*

For a complete graph, r_m has a closed form: sort both batches increasingly and pair the m smallest morning draws with the m smallest afternoon draws in opposite order, $r_m = \sum_{i=1}^{m} (a_{(i)} + b_{(m+1-i)} - B)_+$ (a classical exchange argument, Appendix A). For a general graph, r_m is a minimum-cost matching of cardinality m.

A worked example. Take morning draws $(5, 15)$, afternoon draws $(14, 15)$, capacity $B = 20$, $K = 15$, $H = 49$, $f = 7$, and the complete graph. Then $(r_0, r_1, r_2) = (0, 0, 9)$: the two-bus crossed pairing needs 9 kWh of midday charge, while pairing by index would need 10. With $F(E, L) = 4E + (E^2 + L^2)/20$, the physical optimum is the two-bus plan at $E = 9$, with $D = 2681/20$. The hull optimum lies at $E^* = 151/18$ on the segment joining the three-bus plan at $E = 0$ to the two-bus plan at $E = 9$, with $\mathrm{CH} = 434201/3240$ and $\Delta = 121/3240$ (Figure 1a). At its own marginal prices the physical optimum has regret $11/10$, about thirty times Δ: the support failure costs the operator far more in bill terms than the planning gap costs the system.

5 Price formation on the exact class

We now use the exact description to answer the two questions from the introduction for the lossless class (3). The same arguments apply to the lossless bidirectional class, whose layers are intervals $[t_m, U]$ with $t_m = \max\{S_0 - H,\ S_0 - N_m B,\ 2r_m - K\}$ (Appendix A).

On the physical set, the terminal load is $S_0 - E$. A posted price $p = (p_E, p_L)$ therefore charges $p_L S_0 + \pi E$ with $\pi = p_E - p_L$, and the operator's response depends only on the *spread* π. Assume $\mathrm{Lo} < U$, and let F be convex and differentiable on a neighbourhood of the attainable loads. Let let $g(E) = \partial_E F - \partial_L F$ at $(E, S_0 - E)$. This is the marginal-cost spread at early load E, and it equals $G'(E)$. Assume g is continuous and strictly increasing on $[\mathrm{Lo}, U]$, let E^* be the unique hull optimum, and let $\pi^* = g(E^*)$, the *convex-hull spread*. The response correspondence $R(\pi)$ is the set of physical loads minimizing $fN_m + \pi E$. We use vertex selections, which pick $R(\pi)$ among layer endpoints, as a linear-programming solver would.

Anatomy of the gap. A positive gap needs a sloping face that is expensive enough to straddle, and its size is controlled by the face length.

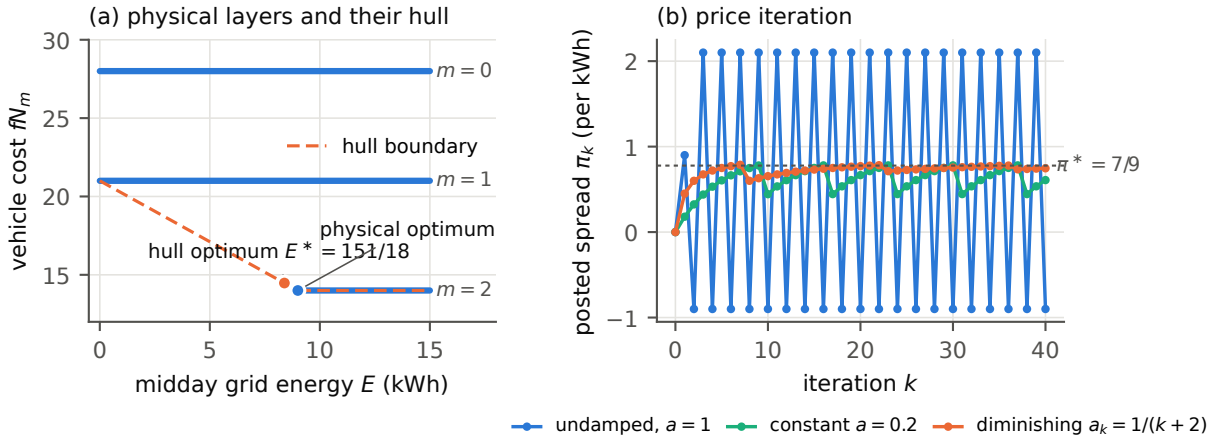


Figure 1: The worked example. (a) Physical layers (solid) and the lower hull boundary (dashed); the hull optimum lies strictly inside the sloping face, so $\Delta > 0$. (b) Price iteration on the scalar spread $\pi = p_E - p_L$: undamped prices enter a 2-cycle that alternates between loads 0 and 15. A constant step $a = 0.2$ keeps prices in a band around the convex-hull spread $\pi^* = 7/9$ while loads alternate between 0 and 9. Diminishing steps converge to π^*.

Proposition 6 (Gap conditions)**.** *Let* $\eta_{\max} = \max_{m_0 < m \le m_{\max}} \eta_m$ *(zero if there is no sloping face).*

(i) If $\Delta > 0$*, then* E^* *lies strictly between consecutive breakpoints* $u < u'$ *with* $\pi^* = f/(u' - u)$*. In particular* $\max_{[\mathrm{Lo},U]} g \ge f/\eta_{\max}$*: no gap can occur when the vehicle cost exceeds* $\eta_{\max}$ *times the largest marginal-cost spread.*

(ii) If g *is* κ*-Lipschitz, then* $\Delta \le \kappa(u' - u)^2/8 \le \kappa\eta_{\max}^2/8$.

Part (i) has a plain reading. Straddling a face means preferring a mixture of "one more pair, $u' - u$ more kWh in the early window" and "one fewer pair". That is worthwhile only when the marginal spread prices the forced energy at the vehicle cost it saves. Part (ii) says the gap, unlike the regret, is of second order in the face length.

Price iteration. A natural way to find a supporting price is to post a price, observe the fleet's response, and move the price toward the marginal cost of that response:

$$\pi_{k+1} = (1 - a_k)\pi_k + a_k\, g(E_k), \qquad E_k \in R(\pi_k), \quad a_k \in (0, 1]. \tag{4}$$

Applied componentwise to (p_E, p_L), the damped update $p_{k+1} = (1 - a_k)p_k + a_k \nabla F(e_k)$ induces exactly (4) on the spread. It is the natural damped fixed-point scheme for computing a price that equals the marginal cost of the response it induces. A fixed point is a spread with $\pi = g(E)$ for some selected $E \in R(\pi)$, that is, a supported plan. The behaviour hinges on whether E^* is a strict vertex of the hull boundary or lies in the relative interior of a maximal face $[w, w']$, which may be a sloping face (possibly containing further breakpoints) or the final flat one.

Lemma 7 (Sign property)**.** *For every* π *and every* $E \in R(\pi)$, $(\pi - \pi^*)(g(E) - \pi^*) \le 0$*. Consequently, for every step* $a \in (0, 1]$, $|\pi_{k+1} - \pi^*| \le \max\{(1 - a)|\pi_k - \pi^*|, aM\}$*, where* $M = \max_{E \in [\mathrm{Lo},U]} |g(E) - \pi^*|$.

Theorem 8 (Price iteration on the two-batch class)**.** *Under the standing assumptions of this section:*

(a) Fixed points. *Some vertex selection has a fixed point if and only if E^* is a breakpoint or $E^* = U$. In particular there is no fixed point when $\Delta > 0$, nor when E^* lies inside the flat face, although $\Delta = 0$ in the latter case.*

(b) Undamped iteration *($a_k \equiv 1$, a deterministic selection). Every orbit is eventually periodic with period one or two. If E^* lies in the relative interior of a maximal face $[w, w']$ and $\pi_0 \neq \pi^*$, the period is two and the cycle's loads satisfy $E_{\mathrm{lo}} \leq w < w' \leq E_{\mathrm{hi}}$.*

(c) Constant step *$a \in (0, 1)$. Every orbit enters the interval $|\pi - \pi^*| \leq aM$ after finitely many steps and remains there. If E^* is a strict vertex in the interior of its normal cone, and aM is smaller than the distance from π^* to the cone's boundary, then $\pi_k \to \pi^*$ geometrically and $E_k = E^*$ eventually. If E^* lies in the relative interior of a maximal face $[w, w']$, the prices converge for at most countably many initial prices, and for none if E^* is not a breakpoint. If moreover aM is smaller than the distance from π^* to the far ends of the normal cones of w and w', then eventually every load lies in $[w, w']$ and equals w or w' unless $\pi_k = \pi^*$ exactly; unless the orbit reaches π^* and the selection then picks E^*, both w and w' recur infinitely often; and $\pi_k \in [\pi^* - a\, d_-,\ \pi^* + a\, d_+]$ with $d_- = \pi^* - g(w)$ and $d_+ = g(w') - \pi^*$.*

(d) Diminishing steps *($a_k \to 0$, $\sum_k a_k = \infty$). Then $\pi_k \to \pi^*$, and the step-weighted averages of $g(E_k)$ converge to π^*. If g is affine, the step-weighted average load converges to E^*.*

Lemma 7 is the whole engine. It follows from monotonicity of the hull boundary's subdifferential, and the proofs are in Appendix C. Figure 1b illustrates all three regimes on the worked example. The step-size conditions in (c) matter. In the worked example with the steeper supply $F(E, L) = 16E + (E^2 + L^2)/4$, the hull optimum is the supported vertex $E^* = 9$, yet undamped iteration from $\pi_0 = 3$ alternates between loads 0 and 15 forever, while the step $a = 1/10$ converges. Part (b) also assumes a selection that depends only on the current price; history-dependent tie rules are outside its scope.

What the theorem says about cycling. With small steps, non-convergence of (4) is caused by a *degenerate tangency*: the optimal load sits on a face of the frontier, where the linear response jumps between the face's endpoints. That happens in two different situations. On a sloping face the interior loads are not physical, so $\Delta > 0$ and no posted price can support any plan; side payments or schedule mixing over days are unavoidable. On the flat face (the cheapest fleet has spare midday capacity) every interior load is physical and supported at π^*, but a vertex-returning solver never selects it. A tie-breaking rule or a quantity signal is then enough. Damping cannot tell these apart, and neither can the presence of a cycle. The gap Δ, or equivalently whether E^* is physical, is the right diagnostic. The calibrated study in Section 7 shows that the second situation is the common one.

What averaging buys. Parts (c) and (d) give the hull a physical meaning. Iterating with small steps and operating the visited schedules on successive days implements, on average, the convex-hull plan. This is the one-dimensional counterpart of ergodic primal recovery in dual subgradient methods [13, 17], here with exact identification of which two schedules alternate.

6 Where the reduction stops

The reduction rests on one fact: at fixed cardinality, total energy and every aggregate bound except the mandatory purchase are independent of the matching. We now show which changes preserve that fact and which destroy it.

Common transition energy preserves it. If every matched pair performs the same compulsory task of energy $d \geq 0$ during a noncharging interval between the early window and the B batch, then total consumption is $S_0 + md$, the paired interval becomes $[a_i + b_j + d - B, a_i]$, and edges with $b_j + d > B$ are removed. Theorem 3 holds with these substitutions (Appendix D).

Edge-dependent transition energy breaks it. When the transition energy depends on the pair, total energy depends on the matching, and a minimum-requirement matching no longer dominates.

Proposition 9 (Support failure on a complete assignment graph)**.** *For every $k \geq 1$ there is a charge-only two-batch instance with $2k$ services per batch, the complete assignment graph, pair-dependent compulsory transition energies, and a fleet cap of $2k$, in which every perfect matching is physically feasible and, for the strongly convex-in-E supply $F(E, L) = 3E + L + (E-3)^2/(2K)$ and any $f \geq 0$,*

$$D = \mathrm{CH} + \frac{9}{2K} > \mathrm{CH}.$$

The construction and proof are in Appendix D. Every assignment alternative is admitted, so the gap is not an artifact of a sparse compatibility graph or of infeasible pairings. The gap shrinks as K grows; the statement is an existence result, not a prevalence claim.

A third batch makes feasibility hard.

Proposition 10 (Three batches)**.** *With three successive batches of n simultaneous services, complete zero-energy compatibility between consecutive batches, no charging before the last batch, fleet cap n and an adequate terminal connector, deciding whether a feasible fleet exists is strongly NP-complete.*

The proof reduces from numerical three-dimensional matching [8, problem SP16] (Appendix D). Pairwise matching therefore cannot describe longer duties, which is why the exact class stops at two batches.

Positive setup time. The construction in Theorem 3 uses arbitrarily many short neutral cycles and zero switching time. A positive setup time per action breaks exactness: it can make a layer infeasible while neighbouring layers stay feasible, and the number of buses needing early charge then matters separately from r_m. A conservative inner approximation with explicit slack conditions is possible; we leave a systematic treatment to future work.

Beyond the class: one certified public timetable. To show that the obstruction is not an artifact of synchronized batches, we certified a strictly positive gap on the 37-service Hildenbrand timetable from the Sistig et al. dataset, using a single depot with its complete deadhead matrix, varied service times, and one shared 360 kW connector, 400 kWh batteries and synthetic economics

(vehicle cost 100, $F(e) = \sum_{t=1}^{30}(0.2e_t + e_t^2/1800)$ over hourly intervals). A feasible two-plan mixture bounds the hull from above. An exhaustive rational branch-and-bound certificate, checked independently, bounds every physical fleet from below:

$$\mathrm{CH} \le 441.142238 < 441.532912 \le D \le 441.532913.$$

By Proposition 2, no physical schedule is supported at its own marginal prices. The result holds under two separately certified energy models: the dataset's stored consumption values on the declared movement graph, and an exactly additive law covering arbitrary finite walks under triangle-inequality premises. Appendix E summarizes the certificate and its scope. The gap is about 0.09% of D, consistent with the magnitudes in Section 7.

7 A calibrated study with public timetables

The theory says when support can fail and how large the gap can be. We now ask whether those conditions arise for real service energies.

Data and instance construction. The dataset of Sistig et al. [21] (CC BY 4.0) contains timetables, stops and directed deadhead matrices for German bus operators. For each of the 20 operators in the archive we take the weekday day type with the most service trips. Trips departing in the morning peak [05:30, 09:30) form the A batch and trips departing in the afternoon peak [14:30, 18:30) form the B batch. Within each batch we chain trips into the minimum number of vehicle runs (maximum bipartite matching on the reachability graph). Each run's energy uses the source study's vehicle parameters: 1.2 kWh/km in service, 0.96 kWh/km deadheading, and 16 kWh/h of auxiliary load away from the depot, including pull-out and pull-in to the nearest depot [22]. We drop 108 trip records whose implied speed exceeds 90 km/h as data errors, and the two operators with fewer than five runs per batch. This leaves 18 operators with 6–413 morning runs and 6–298 afternoon runs. The 3913 run energies have median 154 kWh and interdecile range 110–204 kWh. The resulting instances keep the real heterogeneity of peak-run energies and nothing else: off-peak service, crews and multi-depot structure are not modeled.

Economics. We treat economics as a sensitivity grid rather than a calibration. The daily vehicle cost is $f = 150$ EUR in the sweeps; critical spreads scale linearly in f. The supply cost in each window is quadratic, $\bar{p}_w E + \beta E^2/2$, so the marginal-cost spread is $g(E) = s_0 + \beta(2E - S_0)$, which is Lipschitz with $\kappa = 2\beta$. The zero-load spread $s_0 = \bar{p}_E - \bar{p}_L$ ranges over 37 values from -0.2 to 1.6 EUR/kWh. The price impact is $\beta = \rho\,\bar{p}/S_0$ with $\bar{p} = 0.15$ EUR/kWh and $\rho \in \{0.5, 1, 2, 4\}$: drawing the fleet's whole daily energy in one window would raise that window's marginal cost by $\rho\bar{p}$. We report costs relative to the total daily cost, vehicles plus energy, with the overnight zero-load price set to $\bar{p}$. The midday budget K is either ample ($K = P$) or half that, and the overnight budget is ample. The model has no in-service charging, so a run needing more than the usable battery B cannot be served. We exclude such operator–battery combinations rather than trimming their runs (57 runs exceed 250 kWh, 13 exceed 300 kWh, 8 exceed 350 kWh, none exceed 400 kWh).

Result 1: with today's batteries the obstruction is mostly impossible. Proposition 6(i) turns the data into a necessary condition: a gap requires a marginal spread of at least $f/\eta_{\max}$, which depends only on run energies and battery size. Figure 2 shows it per operator. With $B = 400$ kWh, 15 of 18 operators have $r_m = 0$ for every m: every pairing of a morning and an afternoon run needs no midday charging. The frontier is then flat, and support cannot fail at any price. For the

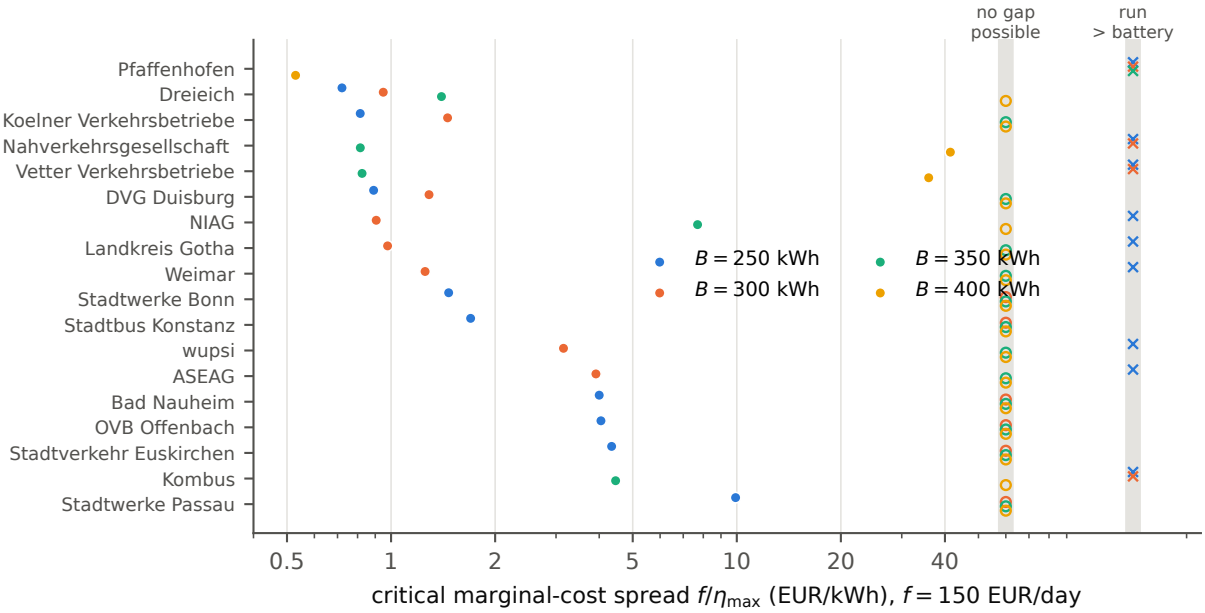


Figure 2: Critical marginal-cost spread $f/\eta_{\max}$ below which no support gap can occur (Proposition 6), for a vehicle cost of 150 EUR/day, per operator and usable battery size B. Hollow markers: $r_m = 0$ for all m, so no gap is possible at any price. Crosses: some peak run needs more than B, so the combination is outside the model.

remaining three the critical spread is 0.53 EUR/kWh or more, and above 35 EUR/kWh for two of them. Smaller batteries make forced charging real. At $B = 350$ kWh, 5 of 17 eligible operators admit a gap, from 0.81 EUR/kWh (median 1.40). At $B = 300$ kWh, 8 of 14 do, from 0.91 EUR/kWh (median 1.27). At $B = 250$ kWh, all 9 eligible operators do, from 0.72 EUR/kWh (median 1.70). These thresholds are several times the 0.15 EUR/kWh reference energy price. Support failure therefore needs capacity or scarcity pricing, or a much cheaper vehicle, together with a battery sized close to the paired energy.

Result 2: gaps are rare and small; regret is not. At $B = 400$ kWh the grid has $18 \times 37 \times 4 = 2664$ scenarios. Only 7 have $\Delta > 0$, all with $s_0 > 0.3$ EUR/kWh, and the largest gap is 0.023% of total daily cost. At $B = 300$ kWh (14 eligible operators, 2072 scenarios) there are 6, with gaps up to 0.054%. In gap scenarios, the planner optimum's regret at its own marginal prices is a median 8.5 times the gap at $B = 400$ (4.2 at $B = 300$). The operator's incentive to deviate is several times the welfare at stake, as in the worked example. Halving the midday budget leaves these gap counts unchanged.

Result 3: cycling is common and mostly unrelated to the gap. Undamped iteration, started from the spread at the smallest attainable early load, cycles in 397 of the 2664 scenarios at $B = 400$ kWh (15%). Of these, 390 have $\Delta = 0$ (Figure 3). Classifying the hull optimum explains them: it lies inside the flat face in 384 cycling scenarios, inside a sloping face in 7 (exactly the gap scenarios) and at a vertex in 6, where the undamped step is too large for part (c) of Theorem 8. Every scenario with a face optimum cycles. The flat-face cases concentrate at small spreads: the cheapest fleet has spare midday capacity, and the vertex response jumps between the two ends of

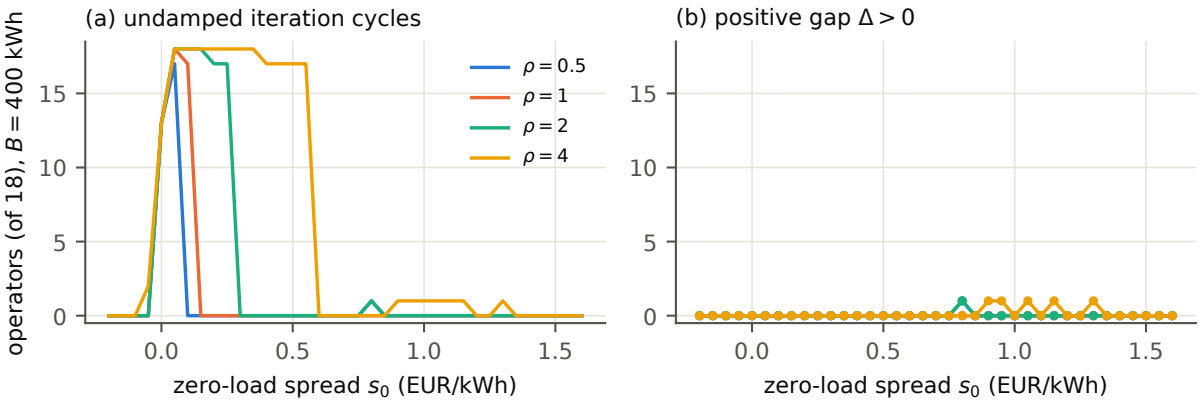


Figure 3: Number of operators (of 18; $B = 400\,\text{kWh}$, $f = 150\,\text{EUR/day}$, ample midday budget) whose undamped price iteration cycles (a) and whose planning problem has a positive gap (b), as functions of the zero-load spread s_0 and the price impact ρ. Cycling concentrates at small spreads with no gap.

its interval. With half the midday budget, 189 scenarios (7%) cycle, 182 of them without a gap. At $B = 300\,\text{kWh}$ the pattern is the same: 307 of 2072 cycle and 301 of these have no gap. A study that counted price cycles as evidence of assignment nonconvexity would overstate it by a factor of about fifty here.

8 Discussion

For tariff design. The exact class isolates the economic content of assignment nonconvexity. A depot facing a marginal-cost tariff will under- or over-charge at midday only when the spread prices forced charging at the vehicle cost it saves. That requires battery capacity close to the paired energy of peak runs, together with a spread near f/η. When it happens, the welfare loss is second order, at most $\kappa\eta^2/8$, but the operator's bill regret is first order. At the convex-hull price an uplift of at most Δ supports the planner optimum (Section 3). That is far cheaper than compensating the regret at the operator's own marginal prices, and it works where re-pricing until convergence does not.

How much the price-maker concern matters. The calibrated study gives a sober answer for peak operations. With current battery sizes, most operators' peak runs pair freely, and the assignment nonconvexity is invisible to any price. Where it is visible, it needs spreads several times the reference energy price, and the welfare at stake is a few hundredths of a percent of daily cost. The operator's incentive to deviate, however, is several times larger. The binding consideration for tariff design is therefore the incentive problem, not the efficiency loss. Battery downsizing, demand charges and scarcity pricing all push depots toward the regime where the obstruction appears, so the threshold $f/\eta_{\max}$ is the quantity to monitor.

For price-coordination algorithms. Theorem 8 separates two causes of non-convergence. Degenerate flat-face responses can be removed by selection rules, such as choosing the response closest to the previous load or announcing a target quantity. Sloping-face gaps cannot be removed by any

price. Constant damping only narrows the band, and an overly large step can make prices cycle even around a supported vertex. Diminishing steps with averaging recover the hull plan. Iterative tariff negotiation between a depot and a supplier should therefore report the gap, or a certificate that E^* is physical, rather than a cycle count.

Limitations. The exact theorem assumes two synchronized batches, common availability and efficiencies, zero-time transitions, ideal switching and period-aggregate settlement. Section 6 shows that edge-dependent transitions and a third batch are genuine boundaries, and that setup time requires separate treatment. The calibrated study uses real service energies but stylized peak runs and illustrative economics. It is not an estimate of savings for any operator. The 37-service certificate establishes existence beyond the class for one timetable with synthetic economics.

Open directions. Three extensions would most increase reach: a characterization for staggered charging windows, where serialization is no longer automatic; an approximate reduction with setup time and its error bound; and an analysis of the strategic operator who anticipates the price change, for which the exact projection is the natural strategy space.

References

[1] Feras Al Taha and Eilyan Bitar. When are lossy energy storage optimization models convex?, 2024. URL https://arxiv.org/html/2403.14010v2. Version 2, 19 November 2024.

[2] Feras Al Taha, Tyrone Vincent, and Eilyan Bitar. An efficient method for quantifying the aggregate flexibility of plug-in electric vehicle populations, 2024. URL https://arxiv.org/abs/2207.07067v3. Version 3, 18 January 2024.

[3] Panagiotis Andrianesis, Dimitris Bertsimas, Michael C. Caramanis, and William W. Hogan. Computation of convex hull prices in electricity markets with non-convexities using Dantzig–Wolfe decomposition. *IEEE Transactions on Power Systems*, 37(4):2578–2589, 2022. doi: 10.1109/TPWRS.2021.3122000.

[4] Hélène Arvis, Olivier Beaude, Nicolas Gast, Stéphane Gaubert, and Bruno Gaujal. Integrating aggregated electric vehicle flexibilities in unit commitment models using submodular optimization. In *2026 European Control Conference (ECC)*, pages 760–766, 2026. URL https://arxiv.org/html/2511.11191v2. Technical comparison uses author preprint version 2, 4 August 2026.

[5] Alicia Blatiak, Federica Bellizio, Luis Badesa, and Goran Strbac. Value of optimal trip and charging scheduling of commercial electric vehicle fleets with vehicle-to-grid in future low inertia systems, 2022. URL https://arxiv.org/abs/2204.11565v1. Author manuscript. Related published DOI: 10.1016/j.segan.2022.100738.

[6] Ralf Borndörfer, Andreas Löbel, Fabian Löbel, and Steffen Weider. Solving the electric bus scheduling problem by an integrated flow and set partitioning approach. In *24th Symposium on Algorithmic Approaches for Transportation Modelling, Optimization, and Systems (ATMOS 2024)*, volume 123 of *Open Access Series in Informatics*, pages 11:1–11:16, 2024. doi: 10.4230/OASIcs.ATMOS.2024.11. URL https://drops.dagstuhl.de/entities/document/10.4230/OASIcs.ATMOS.2024.11.

[7] Marelot H. de Vos, Rolf N. van Lieshout, and Twan Dollevoet. Electric vehicle scheduling in public transit with capacitated charging stations. *Transportation Science*, 58(2):279–294, 2024. doi: 10.1287/trsc.2022.0253.

[8] Michael R. Garey and David S. Johnson. *Computers and Intractability: A Guide to the Theory of NP-Completeness*. W. H. Freeman, San Francisco, 1979.

[9] Paul R. Gribik, William W. Hogan, and Susan L. Pope. Market-clearing electricity prices and energy uplift, 2007. URL https://www.lmpmarketdesign.com/papers/Gribik_Hogan_Pope_Price_Uplift_123107.pdf. Working paper, 31 December 2007.

[10] Kári Hreinsson, Anna Scaglione, Mahnoosh Alizadeh, and Yonghong Chen. New insights from the shapley-folkman lemma on dispatchable demand in energy markets. *IEEE Transactions on Power Systems*, 36(5):4028–4041, 2021. doi: 10.1109/TPWRS.2021.3065913.

[11] Bowen Hua and Ross Baldick. A convex primal formulation for convex hull pricing. *IEEE Transactions on Power Systems*, 32(5):3814–3823, 2017. doi: 10.1109/TPWRS.2016.2637718.

[12] Lukas Hümbs, Alexander Martin, and Lars Schewe. Exploiting complete linear descriptions for decentralized power market problems with integralities. *Mathematical Methods of Operations Research*, 95:451–474, 2022. doi: 10.1007/s00186-022-00775-z.

[13] Torbjörn Larsson, Michael Patriksson, and Ann-Brith Strömberg. Ergodic, primal convergence in dual subgradient schemes for convex programming. *Mathematical Programming*, 86(2):283–312, 1999. doi: 10.1007/s101070050090.

[14] Karan Mukhi and Alessandro Abate. An exact characterisation of flexibility in populations of electric vehicles, 2023. URL https://arxiv.org/abs/2306.16824. Author manuscript.

[15] Karan Mukhi, Georg Loho, and Alessandro Abate. Exact characterization of aggregate flexibility via generalized polymatroids. *IEEE Transactions on Smart Grid*, 17(3):2131–2142, 2026. doi: 10.1109/TSG.2026.3659513. URL https://arxiv.org/abs/2503.23458v2. Technical comparison uses author preprint version 2, 17 June 2025.

[16] Fabian L. Müller, Jácint Szabó, Olle Sundström, and John Lygeros. Aggregation and disaggregation of energetic flexibility from distributed energy resources. *IEEE Transactions on Smart Grid*, 10(2):1205–1214, 2019. doi: 10.1109/TSG.2017.2761439. URL https://arxiv.org/pdf/1705.02815v1. Technical comparison uses author preprint version 1, 8 May 2017.

[17] Angelia Nedić and Asuman Ozdaglar. Approximate primal solutions and rate analysis for dual subgradient methods. *SIAM Journal on Optimization*, 19(4):1757–1780, 2009. doi: 10.1137/070708111.

[18] Richard P. O'Neill, Paul M. Sotkiewicz, Benjamin F. Hobbs, Michael H. Rothkopf, and William R. Stewart. Efficient market-clearing prices in markets with nonconvexities. *European Journal of Operational Research*, 164(1):269–285, 2005. doi: 10.1016/j.ejor.2003.12.011.

[19] Axel Parmentier, Rafael Martinelli, and Thibaut Vidal. Electric vehicle fleets: Scalable route and recharge scheduling through column generation. *Transportation Science*, 57(3):631–646, 2023. doi: 10.1287/trsc.2023.1199.

[20] Dane A. Schiro, Tongxin Zheng, Feng Zhao, and Eugene Litvinov. Convex hull pricing in electricity markets: Formulation, analysis, and implementation challenges. *IEEE Transactions on Power Systems*, 31(5):4068–4075, 2016. doi: 10.1109/TPWRS.2015.2486380.

[21] Hubert Maximilian Sistig, Philipp Sinhuber, Matthias Rogge, and Dirk Uwe Sauer. Dataset for evaluating costs and operations of public bus fleet electrification, 2025. URL https://doi.org/10.6084/m9.figshare.26088190.v1. Version 1; CC BY 4.0.

[22] Hubert Maximilian Sistig, Philipp Sinhuber, Matthias Rogge, and Dirk Uwe Sauer. Evaluating costs and operations of public bus fleet electrification. *npj Sustainable Mobility and Transport*, 2:15, 2025. doi: 10.1038/s44333-025-00030-y.

[23] Ross M. Starr. Quasi-equilibria in markets with non-convex preferences. *Econometrica*, 37(1): 25–38, 1969. doi: 10.2307/1909201.

[24] Weitiao Wu, Yue Lin, Ronghui Liu, and Wenzhou Jin. The multi-depot electric vehicle scheduling problem with power grid characteristics. *Transportation Research Part B: Methodological*, 155:322–347, 2022. doi: 10.1016/j.trb.2021.11.007.

[25] Z. Wu, F. Guo, J. Polak, and G. Strbac. Evaluating grid-interactive electric bus operation and demand response with load management tariff. *Applied Energy*, 255:113798, 2019. doi: 10.1016/j.apenergy.2019.113798. URL https://doi.org/10.1016/j.apenergy.2019.113798.

[26] Xiangyu Zhou, Kun An, and Jan-Dirk Schmöcker. Optimization of charging and discharging schedules for battery electric buses under the V2G environment. *Transportmetrica B: Transport Dynamics*, 13(1):2506689, 2025. doi: 10.1080/21680566.2025.2506689. URL https://repository.kulib.kyoto-u.ac.jp/handle/2433/300766.

[27] Izabela Zoltowska and Jeremy Lin. Optimal charging schedule planning for electric buses using aggregated day-ahead auction bids. *Energies*, 14(16):4727, 2021. doi: 10.3390/en14164727.

A Proofs for Section 4

A.1 Proof of Theorem 3

Fix one allowed matching M with $|M| = m$, fleet $N = N_m$, requirement $R = \sum_{(i,j)\in M}(a_i + b_j - B)_+$ and $T = S_0 - NB$. Let x_v be a bus's net early battery gain. A paired bus serving A_i and B_j holds $B - a_i$ after its morning service; it cannot exceed capacity, so $x_v \le a_i$, and it must hold at least b_j before the afternoon service, so $x_v \ge a_i + b_j - B$. An A-only bus has $x_v \in [a_i - B, a_i]$, and a B-only bus, full before the early window, has $x_v \in [b_j - B, 0]$. These boxes are nonempty because $a_i, b_j \le B$. Their lower endpoints sum to T, their positive lower endpoints to R, and their upper endpoints to P; every upper endpoint is nonnegative.

Necessity. Whatever the order of actions, $x_v = \alpha C_v - X_v/\delta$ for the bus's early purchases C_v and exports X_v, so the total Q lies in $[T, P]$. A positive lower endpoint needs at least its value divided by α in purchases, hence $C \ge R/\alpha$. Exports are nonnegative, the shared connector carries gross energy $C + X \le K$, and full terminal replenishment needs the terminal purchase $L = (S_0 - Q)/\alpha \le H$. Terminal purchases are nonnegative because $Q \le P < S_0$.

Sufficiency. Fix totals (C, X) satisfying these inequalities with R in place of r_m. For any $Q \in [T, P]$ the boxes admit gains whose positive parts sum to $A = \max\{R, Q\}$. If $Q \ge R$, start from the positive lower endpoints and zero elsewhere, and distribute $Q - R$ through the remaining

headroom. If $Q < R$, keep the mandatory positive gains and distribute the negative total $Q - R$ through the negative lower intervals, whose combined lower endpoint is $T - R$. One-direction exchanges then realize $C_0 = A/\alpha$ and $X_0 = \delta(A - Q)$. Since $X \geq 0$ implies $\alpha C \geq Q$, and $\alpha C \geq R$, the residual $z = C - C_0$ is nonnegative and $X - X_0 = \alpha\delta z$.

Realize these residuals by battery-neutral cycles on one bus before its one-direction exchange. At early state of charge $s \in [0, B]$, use charge-then-discharge cycles of depth at most $B - s$ if $s \leq B/2$, and discharge-then-charge cycles of depth at most s otherwise. The permitted depth is at least $B/2 > 0$. Split the residual battery throughput αz into finitely many such chunks. Each returns to s, stays in $[0, B]$, and uses grid purchases depth$/\alpha$ and exports $\delta\cdot$depth, so together they realize exactly $(z, \alpha\delta z)$. Execute the cycles and then all one-direction exchanges serially: their gross energy is $C + X \leq K$, so they fit the common early window at the connector's power. The endpoint boxes guarantee the afternoon service, and nonnegative service draw preserves every intermediate state-of-charge bound. Refill serially at the terminal connector using total $L \leq H$. If $K = 0$ all early actions vanish; if $H = 0$, then $P < S_0$ makes every polygon empty, consistent with infeasibility.

Replacement. At fixed m, the quantities S_0, P, N_m, T and the fleet cost do not depend on the matching; only R does, and only through $C \geq R/\alpha$. Hence the polygon of a matching attaining r_m contains the polygon of every other matching of size m, and every point of it has just been realized. The union over matchings of size m therefore equals the polygon W_m, and the union over m gives $\mathcal{P}$. Each W_m is defined by five half-planes, so it has at most five vertices, and $\operatorname{conv}\mathcal{P}$ is the convex hull of their images. With exports prohibited, the construction above uses no exports when $X = 0$: then $\alpha C = Q \geq R$, so $A = Q$, $X_0 = 0$ and $z = 0$. □

Idle buses. Identical paid idle buses under the total fleet cap $n_A + n_B$ do not change $\mathcal{P}$. An idle bus has the box $[-B, 0]$. A plan with m pairs and q idle buses has $q \leq m$ by the cap; deleting q pairs and serving their services separately preserves N, S_0, P, T and the fleet cost while lowering R, and reproduces the same (C, X).

A.2 The lossless cases

Charge-only. With $\alpha = \delta = 1$ and $X = 0$, W_m becomes $\{C : \ C \geq r_m,\ C \leq K,\ S_0 - N_m B \leq C \leq P,\ C \geq S_0 - H\}$. Since $S_0 - N_m B$ is the sum of all lower endpoints and r_m the sum of their positive parts, $S_0 - N_m B \leq r_m$, and the layer is $\{E : \max\{\mathrm{Lo}, r_m\} \leq E \leq U\}$ with $L = S_0 - E$, which is (3). Feasibility is $\mathrm{Lo} \leq U$ because the empty matching has $r_0 = 0$.

Bidirectional. With $\alpha = \delta = 1$ and exports allowed, $E = C - X$ and $L = S_0 - E$. A point E is attainable in layer m if and only if $\max\{S_0 - N_m B, S_0 - H\} \leq E \leq P$ and some $C \geq \max\{r_m, E\}$ satisfies $2C - E \leq K$, that is, $E \leq K$ and $E \geq 2r_m - K$. The layer is therefore $[t_m, U]$ with $t_m = \max\{S_0 - H,\ S_0 - N_m B,\ 2r_m - K\}$ and $U = \min\{P, K\}$.

A.3 Monotonicity, convexity and the frontier

Deleting an edge from a minimum matching of size m shows $r_{m-1} \leq r_m$. For discrete convexity, consider the unit-capacity source–A–B–sink network on the allowed edges with costs $(a_i + b_j - B)_+$. Its minimum-cost flow LP is integral at integer flow values $m = 0, \ldots, \nu$, and its value is convex in the required flow, so $2r_m \leq r_{m-1} + r_{m+1}$. This uses bipartiteness, not completeness of Γ.

All thresholds at or below Lo are dominated in cost by m_0. The first retained increment is $r_{m_0+1} - \mathrm{Lo} \leq r_{m_0+1} - r_{m_0}$, and later increments are the original nondecreasing matching increments, all positive after m_0. The slopes $-f/\eta_m$ are therefore nondecreasing, followed by the flat segment. Endpoints are physical, interpolants are hull mixtures, and every physical branch lies on or above

this convex graph, so the graph is exactly the lower hull boundary. Consecutive breakpoints differ by one cardinality.

A.4 Proof of Corollary 5

The convex frontier plus strictly convex continuous G has a unique minimizer E^*. At Lo, at a breakpoint or on the flat segment, a physical plan attains the frontier, so $D = \mathrm{CH}$. Strictly between consecutive breakpoints the frontier's operating cost is strictly below that of every physical branch because $f > 0$, so no physical plan attains the hull optimum; compactness of the finite union of physical segments then gives $D > \mathrm{CH}$.

A.5 Sorted matching for the complete graph

Replacing a selected service by an unused smaller draw while keeping its partner cannot increase the nondecreasing edge cost, so some optimal size-m matching uses the m smallest draws in each batch. With $h(t) = (t - B)_+$ and $a \le a'$, $b \le b'$, convexity gives $h(a+b) + h(a'+b') \ge h(a+b') + h(a'+b)$. Swapping any two partners ordered in the same direction therefore cannot raise the cost, and repeated exchanges yield opposite order. For a non-complete graph this shortcut fails: with A draws $(5, 15)$, B draws $(14, 15)$ and $B = 20$, allowing only the index-matched edges gives $r_2 = 10$ instead of 9.

A.6 The worked example

The requirement matrix is $\left(\begin{smallmatrix} 0 & 0 \\ 9 & 10 \end{smallmatrix}\right)$, so $r_0 = r_1 = 0$, $r_2 = 9$, $S_0 = 49$, Lo $= 0$ and $U = 15$. With $G(E) = 2401/20 - 9E/10 + E^2/10$, the physical minima of layers $m = 0, 1$ are at $E = 9/2$, with values $5841/40$ and $5561/40$; layer $m = 2$ is minimized at its endpoint $E = 9$ with value $2681/20$. On the sloping hull face the objective is $2821/20 - 151E/90 + E^2/10$, minimized at $E^* = 151/18$ with value $434201/3240$, so $\Delta = 121/3240$. The hull point mixes the three-bus plan at $E = 0$ and the two-bus plan at $E = 9$ with weights $11/162$ and $151/162$. At the physical optimum's marginal prices $(49/10, 4)$ each layer's bill is minimized at its lowest endpoint; bill plus vehicle cost is 224, 217 and $2181/10$, so the regret is $11/10$. Both mixed plans have explicit serial witnesses: at $E = 0$ pair the 5-draw A with the 14-draw B and refill $19, 15, 15$ on the terminal connector; at $E = 9$ the 15-draw A bus charges 9 at midday in the crossed pairing and both buses refill 20.

B Proof of Proposition 2

For any hull point (e, c), linear minimization gives $c + p^\top e \ge V(p)$. If $p \in \partial F(e(x))$, then $F(e) \ge F(e(x)) + p^\top(e - e(x))$. Adding and minimizing over the hull gives $\mathrm{CH} \ge V(p) + F(e(x)) - p^\top e(x) = c(x) + F(e(x)) - r(x; p)$, which is the regret bound; if $r(x; p) = 0$, then $D = \mathrm{CH}$.

Conversely let $D = \mathrm{CH}$. Compactness gives a physical minimizer x. The relative-interior condition gives strong Fenchel duality with an attained maximizer, $\mathrm{CH} = V(p^*) - F^*(p^*)$. Then

$$\underbrace{c(x) + (p^*)^\top e(x) - V(p^*)}_{\ge 0} + \underbrace{F(e(x)) - (p^*)^\top e(x) + F^*(p^*)}_{\ge 0} = D - \mathrm{CH} = 0,$$

so both terms vanish: Fenchel equality gives $p^* \in \partial F(e(x))$, and the first term gives $r(x; p^*) = 0$. The same identity holds for any physical optimum and any dual maximizer when $D > \mathrm{CH}$, with right-hand side Δ; since both terms are nonnegative, $r(x; p^*) \le \Delta$. □

C Proofs for Section 5

Throughout, $\bar{c}$ is the lower hull boundary on $[\mathrm{Lo}, U]$, extended by $+\infty$ outside, and $H(\pi) = \arg\min_E\{\bar{c}(E) + \pi E\}$ is the hull response. The finite candidate set $\mathcal{V}$ consists of all layer endpoints (ℓ_m, fN_m) and (U, fN_m). Since a linear function attains its minimum over $\operatorname{conv}\mathcal{P}$ at a point of $\mathcal{V}$, every $E \in R(\pi)$ satisfies $\bar{c}(E) = fN_m$ for its layer and $E \in H(\pi)$. The layer endpoints that lie on $\bar{c}$ are exactly the breakpoints Lo, r_m $(m_0 < m \le m_{\max})$ and the point U. The hull optimum satisfies $-\pi^* \in \partial\bar{c}(E^*)$ because G is differentiable.

C.1 Proof of Proposition 6

(i) By Corollary 5, $\Delta > 0$ places E^* strictly between consecutive breakpoints $u < u'$, where $\bar{c}$ is affine with slope $-f/(u'-u)$ because consecutive breakpoints differ by one cardinality. Optimality at an interior point gives $g(E^*) = f/(u'-u) \ge f/\eta_{\max}$, and $\max g \ge g(E^*)$.

(ii) On $[u, u']$ let $\phi = \bar{c} + G$. Then $\phi'(E) = -f/(u'-u) + g(E)$ is κ-Lipschitz and vanishes at E^*, so $\phi(E) - \phi(E^*) \le \kappa(E - E^*)^2/2$. Both u and u' are physical with cost $\bar{c}$, so $D \le \min\{\phi(u), \phi(u')\}$ while $\mathrm{CH} = \phi(E^*)$. One of u, u' is within $(u'-u)/2$ of E^*, which gives $\Delta \le \kappa(u'-u)^2/8$. □

C.2 Proof of Lemma 7

Let $E \in R(\pi) \subseteq H(\pi)$, so $-\pi \in \partial\bar{c}(E)$, and recall $-\pi^* \in \partial\bar{c}(E^*)$. Monotonicity of the subdifferential of the convex function $\bar{c}$ gives $(-\pi + \pi^*)(E - E^*) \ge 0$, that is, $(\pi - \pi^*)(E - E^*) \le 0$. Since g is increasing and $g(E^*) = \pi^*$, $(\pi - \pi^*)(g(E) - \pi^*) \le 0$. For the second claim write $\pi_{k+1} - \pi^* = (1-a)(\pi_k - \pi^*) + a(g(E_k) - \pi^*)$. The two terms have opposite (weak) signs, and for real x, y with $xy \le 0$, $|x + y| \le \max\{|x|, |y|\}$. Since $E_k \in [\mathrm{Lo}, U]$, $|g(E_k) - \pi^*| \le M$. □

C.3 Proof of Theorem 8

We use three facts. (F1) The same monotonicity argument shows that $\pi < \pi'$, $E \in H(\pi)$ and $E' \in H(\pi')$ imply $E \ge E'$. (F2) If E^* lies in the relative interior of a maximal face $[w, w']$, then $\partial\bar{c}(E^*) = \{-\pi^*\}$, the face has slope $-\pi^*$, $[w, w'] \subseteq H(\pi^*)$, and $E^* \in H(\pi)$ only for $\pi = \pi^*$. By (F1), $H(\pi) \subseteq [\mathrm{Lo}, w]$ for $\pi > \pi^*$ and $H(\pi) \subseteq [w', U]$ for $\pi < \pi^*$. (F3) A fixed point $\hat{\pi} = g(\hat{E})$ with $\hat{E} \in R(\hat{\pi})$ has $-g(\hat{E}) \in \partial\bar{c}(\hat{E})$, so $\hat{E}$ minimizes the convex function $\bar{c} + G$; by uniqueness $\hat{E} = E^*$ and $\hat{\pi} = \pi^*$.

(a) By (F3), a fixed point requires $E^* \in R(\pi^*)$ to be a selectable layer endpoint on $\bar{c}$, that is, a breakpoint or U. Conversely, if E^* is a breakpoint or U, then $E^* \in H(\pi^*)$ is a candidate attaining the minimum, so $E^* \in R(\pi^*)$ and a selection choosing it makes π^* a fixed point. If $\Delta > 0$, E^* is not physical and hence not a breakpoint; if E^* lies inside the open flat face $(\ell_{m_{\max}}, U)$ it is physical but neither a breakpoint nor U.

(b) Let σ be the selection, a function of π with values in $R(\pi)$; by (F1) it is nonincreasing, so $h = g \circ \sigma$ is nonincreasing and $h \circ h$ is nondecreasing. For $k \ge 1$, π_k lies in the finite set $g(\{E : (E, c) \in \mathcal{V}\})$. The odd and even subsequences are orbits of the nondecreasing map $h \circ h$, hence monotone, hence eventually constant in a finite set. Thus $\pi_{k+2} = \pi_k$ eventually. In the face case with $\pi_0 \ne \pi^*$, we show by induction that $\pi_k \ne \pi^*$: if $\pi_k \ne \pi^*$ then by (F2) $E_k \ne E^*$, so $\pi_{k+1} = g(E_k) \ne \pi^*$. By (F3) there is then no fixed point along the orbit, and the period is two. If $\pi_a > \pi^*$ is one state of the cycle, (F2) gives $E_a \le w < E^*$, hence $\pi_b = g(E_a) < \pi^*$ and $E_b \ge w'$.

(c) By Lemma 7, $d_k = |\pi_k - \pi^*|$ satisfies $d_{k+1} \le \max\{(1-a)d_k, aM\}$, so $d_k \le \max\{(1-a)^k d_0, aM\}$ and the interval $d \le aM$ is reached in finitely many steps and is invariant.

Vertex case. If π^* is interior to the normal cone $\{\pi : E^* \in H(\pi)\}$ and aM is less than its distance to the cone's boundary, then inside the invariant interval $H(\pi) = \{E^*\}$, so $E_k = E^*$ and $\pi_{k+1} - \pi^* = (1-a)(\pi_k - \pi^*)$.

Face case: non-convergence. Suppose $\pi_k \to \bar{\pi}$. Then $a(g(E_k) - \pi_k) = \pi_{k+1} - \pi_k \to 0$, so $g(E_k) \to \bar{\pi}$. Since E_k ranges over a finite set and g is injective, $E_k = \bar{E}$ for all large k, with $\bar{\pi} = g(\bar{E})$. The set $\{\pi : \bar{E} \in R(\pi)\}$ is closed, being defined by finitely many weak linear inequalities, so $\bar{E} \in R(\bar{\pi})$, and (F3) gives $\bar{E} = E^*$ and $\bar{\pi} = \pi^*$. If E^* is not a breakpoint it is never selected, which is a contradiction. If it is a breakpoint, $E^* \in R(\pi_k)$ forces $\pi_k = \pi^*$ by (F2), so the orbit hits π^* exactly. For a fixed sequence of selected loads, π_k is affine in π_0 with slope $(1-a)^k \neq 0$, so each finite selection sequence admits at most one such π_0, and there are countably many sequences.

Face case: trapping. Write $[\pi^*, \sigma]$ for the normal cone of w and $[\sigma', \pi^*]$ for that of w' (with $\sigma = +\infty$ if $w = \mathrm{Lo}$ and $\sigma' = -\infty$ if $w' = U$), and assume $aM < \min\{\sigma - \pi^*, \pi^* - \sigma'\}$. Inside the invariant interval, $\pi > \pi^*$ forces $E = w$ and $\pi < \pi^*$ forces $E = w'$; at $\pi = \pi^*$ the selection is a breakpoint in $[w, w']$. With $d_- = \pi^* - g(w) > 0$ and $d_+ = g(w') - \pi^* > 0$, an update from $\pi > \pi^*$ gives $\pi' = (1-a)\pi + ag(w) \in [\pi^* - ad_-, \pi)$, and symmetrically from below; from π^* the update stays in $[\pi^* - ad_-, \pi^* + ad_+]$. This band is therefore invariant. Above π^*, prices contract geometrically toward $g(w) < \pi^*$ and cross π^* in finitely many steps; below, they contract toward $g(w') > \pi^*$. Hence the band is entered, and both w and w' recur infinitely often unless some $\pi_k = \pi^*$ and the selection picks E^* (possible only if E^* is a breakpoint), after which the orbit is fixed. Because $\mathrm{Lo} < U$ and g is strictly increasing, $M > 0$, so the interval $d \leq aM$ is reached in finitely many steps.

(d) With $d_{k+1} \leq \max\{(1-a_k)d_k, a_k M\}$, fix $\varepsilon > 0$ and k_ε with $a_k M < \varepsilon$ for $k \geq k_\varepsilon$. After k_ε, the interval $d \leq \varepsilon$ is invariant, and outside it d contracts by the factors $1 - a_k$, whose product diverges to zero because $\sum a_k = \infty$. Thus $\limsup d_k \leq \varepsilon$ for every ε. Summing the update gives $\sum_{k<n} a_k(g(E_k) - \pi_k) = \pi_n - \pi_0$, which is bounded; dividing by $A_n = \sum_{k<n} a_k \to \infty$ and using $\pi_k \to \pi^*$ (Stolz–Cesàro) gives $\sum_{k<n} a_k g(E_k)/A_n \to \pi^*$. If g is affine, this average equals $g(\sum_{k<n} a_k E_k / A_n)$, and invertibility of g gives the load statement. □

Numerical verification. The repository contains an exact rational-arithmetic implementation of the projection, hull, gap and iteration (`code/twobatch.py`) and randomized checks of every statement in this section on several thousand instances: the period and bracketing claims in (b); the absence of fixed points; trapping, band and recurrence in (c) at half the stated step threshold; convergence in (d); and both parts of Proposition 6. Instances are drawn so that the hull optimum falls on sloping faces, flat faces and vertices.

D Proofs for Section 6

D.1 Common transition energy

Suppose every matched pair performs a compulsory task of energy $d \geq 0$, with nonnegative draw, during a noncharging interval after the early window and before the B batch. A paired bus then needs $x_v \geq a_i + b_j + d - B$ to finish its afternoon service, cannot exceed $x_v \leq a_i$, and needs terminal refill $a_i + b_j + d - x_v$. The box is nonempty exactly when $b_j + d \leq B$, so edges violating this are removed. Lower endpoints sum to $S_m - N_m B$ with $S_m = S_0 + md$, positive lower endpoints to the matching's requirement with d included, and the upper endpoints still to P. Because S_m depends only on m, the proof of Theorem 3 applies verbatim with S_0 replaced by S_m.

D.2 Proof of Proposition 9

A disjoint-block construction. Let $w_i = 2^i$ for $i = 0, \dots, k-1$, $W = \sum_i w_i$, $B = 20 \max_i w_i$. Block i has two A and two B services with draws $(B/2+3w_i,\ w_i)$ in each batch. Use windows A $[0,1]$, early charging $[1,2]$, noncharging transition tasks $[2,3]$, B $[3,4]$ and terminal charging $[4,5]$; one early connector of budget $K = kB/2 + 4W$ and one terminal connector. Within a block, the identity edges carry transition energy 0 and the cross edges $6w_i$. The fleet cap is $2k$, so exactly $2k$ buses are used and every bus serves one service from each batch. Identity requires early purchase $6w_i$ and consumes $B + 8w_i$; cross requires none and consumes $B + 20w_i$. For a set I of identity blocks, with $R = 6\sum_{i\in I} w_i$ and $S_{\max} = kB + 20W$, total consumption is $S_{\max} - 2R$, the per-bus early intervals sum to $[R, K]$, and the complete branch is $\{(E,\ S_{\max} - 2R - E,\ 2kf) : R \le E \le K\}$, each point realized by serial charging.

Complete graph. Now admit every A–B edge. Keep the within-block transition energies, and on each edge between different blocks set $d_{ij} = B - b_j$; raise the terminal budget to $H = 2kB$. For any edge put $e_{ij} = a_i + b_j + d_{ij}$ and $r_{ij} = (e_{ij} - B)_+$. Its early purchase interval is $[r_{ij}, a_i]$ whenever $b_j + d_{ij} \le B$, which holds for within-block edges because $B \ge 20 \max w_i$ and with equality between blocks. In the latter case $e_{ij} = a_i + B$ and $r_{ij} = a_i$: the bus refills completely at midday. For a perfect matching M these intervals sum to $[R_M, K]$ with $R_M = \sum_M r_{ij}$, total consumption is $S_M = \sum_M e_{ij}$, and the branch is $L = S_M - E$, $R_M \le E \le K$. Each terminal purchase is at most $e_{ij} - r_{ij} \le B$, so $H = 2kB$ accommodates it; $E = K$ is feasible for every perfect matching.

Give each high service (draw $B/2+3w_i$) in block i the potential $B/2+9w_i$ and each low service the potential w_i, as row potentials u_i and column potentials v_j; their total is $S_{\max}$. Every within-block edge satisfies $e_{ij} + 2r_{ij} = u_i + v_j$. A between-block edge has $e_{ij} + 2r_{ij} = B + 3a_i$, with slack over $u_i + v_j$ equal to $3B/2 - 9w_j$, $2B - w_j$, $B/2 + 2w_i - 9w_j$ or $B + 2w_i - w_j$ for the four high/low combinations. Each is at least $\gamma = w_{\max} + 2w_{\min} \ge 3$, the low-to-high case being limiting because $B/2 \ge 10w_{\max}$. Summing this assignment-potential certificate over M gives $S_M + 2R_M \ge S_{\max}$, with surplus at least γ if any between-block edge is used. Hence every physical point, and every mixture, satisfies $3E + L = S_M + 2E \ge S_M + 2R_M \ge S_{\max}$.

With $F(E, L) = 3E + L + (E-3)^2/(2K)$, total cost is at least $2kf + S_{\max}$. The equal mixture of the within-block corners $R = 0$ and $R = 6$ attains it at load $(3, S_{\max} - 9)$, so $\mathrm{CH} = S_{\max} + 2kf$. On a within-block branch the cost is $2kf + S_{\max} + 2(E - R) + (E-3)^2/(2K)$, increasing in E (derivative at least $2 - 3/K > 0$ since $K \ge 14$), hence minimized at $E = R$; the values $R = 6t$ nearest to 3 are 0 and 6, giving excess $9/(2K)$. Every between-block schedule has excess at least $\gamma \ge 3 > 9/(2K)$. Therefore $D = \mathrm{CH} + 9/(2K)$. □

D.3 Proof of Proposition 10

Reduce from numerical three-dimensional matching, strongly NP-complete [8, SP16]: three multisets of n positive integers must be partitioned into triples, one from each, each summing to B. We may assume total nB and every number at most B, mapping violations to a fixed NO instance. Make the multisets services in $[0,1)$, $[1,2)$ and $[2,3)$ with those numbers as battery draws, with charging unavailable before time 3. Overlap and the fleet cap force exactly n buses, each serving one service per batch; a duty (a, b, c) is feasible exactly when $a + b + c \le B$, and the total nB forces equality. Conversely, a matching yields duties ending empty, which refill serially for B time units each during $[3, 3+nB]$. The construction is polynomial and preserves strong hardness; membership in NP is immediate. □

E The certified public timetable

The case uses the Hildenbrand timetable of Sistig et al. [21]: 37 mandatory services, the complete directed 5×5 deadhead matrix, and depot 15 as the sole charging and parking location. Service energy is 1.2 kWh/km plus 16 kWh per service hour; deadheads use 0.96 kWh/km plus 16 kWh per hour; off-depot waiting uses 16 kWh/h. Each used bus has 400 kWh usable energy and 360 kW acceptance; one 360 kW connector is shared; every bus starts full and returns full by 30:00. The used-bus cost is 100 and the supply cost is $F(e) = \sum_{t=1}^{30}(0.2e_t + e_t^2/1800)$ over hourly intervals, with the exact binary values of these coefficients used throughout.

Upper bound on CH. An incumbent physical plan and a second admitted physical response, which has a higher planning cost but a lower bill at the incumbent's marginal prices, are mixed. Exact quadratic evaluation and one-dimensional optimality conditions give CH ≤ 441.142238.

Lower bound on D. A minimum-fleet argument excludes one bus. For three or more buses, vehicle cost plus the linear mandatory-service energy term already exceeds 477.82. The two-bus domain is covered by a binary branching tree over movement variables. At each leaf, exact tangent planes, signed row multipliers and finite-box residual compensation give a valid rational lower bound, and infeasible leaves carry rational infeasibility certificates. An independent reconstruction checks coverage and every leaf, giving $D \geq 441.532912$. A feasible schedule gives $D \leq 441.532913$.

Scope. The first certificate concerns the declared movement graph (pull-out, direct, via-depot and pull-in modes with fixed leg times). A second, separately checked certificate uses an exactly additive energy law and, under directed triangle inequalities that the source tables satisfy, extends the result to arbitrary finite walks over the location pairs with continuous waiting. The displayed bounds coincide, but neither certificate is transferred to the other model. The gap holds for every vehicle cost $f \geq 100$, since both mixture components and the two-bus lower bound shift equally. Standalone checkers using only the Python standard library re-verify each certificate in under a minute (Appendix F).

F Reproducibility

All code and derived data are in the repository at `paper/draft1/`:

- `code/twobatch.py`: exact projection, hull, gap and price iteration; `code/test_twobatch.py`: the randomized checks of Section 5.
- `code/calibrate.py`: builds peak-run energies from the archive of Sistig et al. [21], written to `data/peak_runs.json`.
- `code/study.py`: the sweep, written to `results/summary.json` and `results/study.json`; `code/figures.py`: all figures.
- `make.sh` regenerates results, figures and the PDF.

The public-timetable certificates and their portable checkers are in the repository directory `research-20261029/`, and an index binding sources, inputs and commands is `paper/focused-theory-20261029/CERTIFICATE_INDEX.md`.